

NIDS Session Report

AI-Based Network Intrusion Detection System

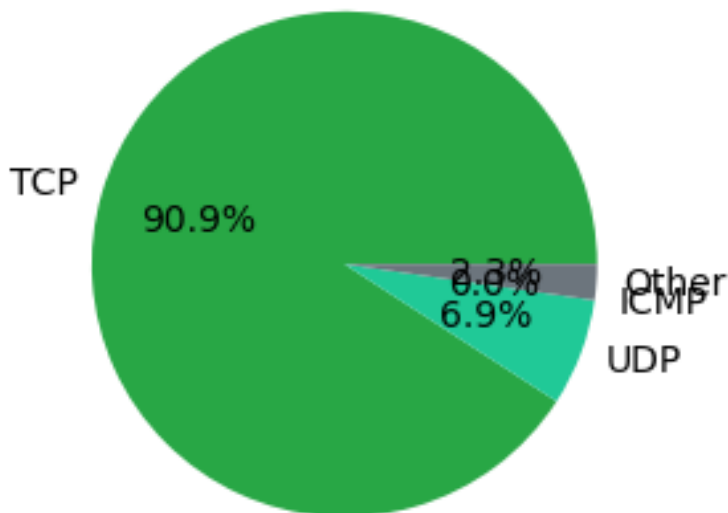
This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Protocol Distribution

Protocol Distribution



Session Summary

Start Time:	1750245437.1010408
Stop Time:	2025-06-18T16:48:25.639427
Uptime:	68 seconds
Packets Processed:	569
Anomalies Detected:	13
Packet Rate:	8.30 packets/sec

Protocol Statistics

TCP PACKETS:	517
UDP PACKETS:	39
ICMP PACKETS:	0
OTHER PACKETS:	13

Security Alerts (5)

Type: Port Scan | Time: 2025-06-18T16:47:19.368465

Details: {'src_ip': '10.0.2.15', 'dst_ip': '10.0.2.3', 'dst_port': 53, 'protocol': 17}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: Ddos Indicator | Time: 2025-06-18T16:47:20.058802

Details: {'src_ip': '34.202.102.234', 'dst_ip': '10.0.2.15', 'packet_size': 1478, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

Type: MI Anomaly | Time: 2025-06-18T16:47:22.146445

Details: {'anomaly_score': -7.301277376559921, 'features': {'packet_count': 97, 'total_bytes': 37531, 'avg_packet_length': np.float64(386.91752577319585), 'std_packet_length': np.float64(1591.474494097518), 'min_packet_length': 54, 'max_packet_length': 14627, 'avg_payload_size': np.float64(329.74226804123714), 'std_payload_size': np.float64(1592.0347175302618), 'total_payload_size': 31985, 'tcp_count': 81, 'udp_count': 12, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.8709677419354839, 'udp_ratio': 0.12903225806451613, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 18, 'unique_dst_ports': 23, 'total_ports_accessed': 26, 'common_port_ratio': 0.3870967741935484, 'high_port_ratio': 0.6129032258064516, 'avg_dst_port': np.float64(32582.677419354837), 'std_dst_port': np.float64(25624.690294970318), 'min_dst_port': 53, 'max_dst_port': 63037, 'unique_connections': 21, 'total_connection_attempts': 93, 'avg_packets_per_connection': 4.428571428571429, 'connection_diversity': 0.21649484536082475, 'max_packets_per_connection': 27, 'unique_src_ips': 11, 'unique_dst_ips': 12, 'src_ip_diversity': 0.1134020618556701, 'dst_ip_diversity': 0.12371134020618557, 'private_src_ip_ratio': 0.18181818181818182, 'private_dst_ip_ratio': 0.16666666666666666, 'packet_rate': 21.30729998313107, 'inter_arrival_time_mean': np.float64(0.04742114990949631), 'inter_arrival_time_std': np.float64(0.1885434733574494), 'burst_count': 56, 'burst_ratio': 0.5833333333333334, 'payload_size_mean': np.float64(329.74226804123714), 'payload_size_std': np.float64(1592.0347175302618), 'zero_payload_ratio': 0.17525773195876287, 'large_payload_ratio': 0.07216494845360824, 'payload_size_entropy': np.float64(0.16551817423413756), 'syn_count': 19, 'fin_count': 0, 'rst_count': 0, 'psh_count': 27, 'ack_count': 74, 'urg_count': 0, 'syn_ratio': 0.2345679012345679, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.3333333333333333, 'ack_ratio': 0.9135802469135802, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:47:22.146426', 'packet_count': 97, 'protocol_distribution': {'tcp': 81, 'udp': 12, 'icmp': 0}, 'connection_info': {'unique_connections': 21, 'unique_src_ips': 11, 'unique_dst_ips': 12}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Port Scan | Time: 2025-06-18T16:48:21.212706

Details: {'src_ip': '10.0.2.15', 'dst_ip': '75.2.76.8', 'dst_port': 443, 'protocol': 6}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: MI Anomaly | Time: 2025-06-18T16:48:22.346289

Details: {'anomaly_score': -7.29958547800657, 'features': {'packet_count': 100, 'total_bytes': 14204, 'avg_packet_length': np.float64(142.04), 'std_packet_length': np.float64(290.507690775993), 'min_packet_length': 54, 'max_packet_length': 1536, 'avg_payload_size': np.float64(86.08), 'std_payload_size': np.float64(291.1846039886038), 'total_payload_size': 8608, 'tcp_count': 95, 'udp_count': 4, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.9595959595959596, 'udp_ratio': 0.04040404040404041, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 11, 'unique_dst_ports': 54, 'total_ports_accessed': 54, 'common_port_ratio': 0.23232323232323232, 'high_port_ratio': 0.7676767676767676, 'avg_dst_port': np.float64(41005.50505050505), 'std_dst_port': np.float64(22349.262516643663), 'min_dst_port': 53, 'max_dst_port': 61356, 'unique_connections': 28, 'total_connection_attempts': 99, 'avg_packets_per_connection': 3.5357142857142856, 'connection_diversity': 0.28, 'max_packets_per_connection': 19, 'unique_src_ips': 22, 'unique_dst_ips': 8, 'src_ip_diversity': 0.22, 'dst_ip_diversity': 0.08, 'private_src_ip_ratio': 0.09090909090909091, 'private_dst_ip_ratio': 0.25, 'packet_rate': 9.618649121724355, 'inter_arrival_time_mean': np.float64(0.10501485159902861), 'inter_arrival_time_std': np.float64(0.2114663380904673), 'burst_count': 65, 'burst_ratio': 0.6565656565656566, 'payload_size_mean': np.float64(86.08), 'payload_size_std': np.float64(291.1846039886038), 'zero_payload_ratio': 0.14, 'large_payload_ratio': 0.04, 'payload_size_entropy': np.float64(0.6536395280563827), 'syn_count': 31, 'fin_count': 4, 'rst_count': 26, 'psh_count': 14, 'ack_count': 90, 'urg_count': 0, 'syn_ratio': 0.3263157894736842, 'fin_ratio': 0.042105263157894736, 'rst_ratio': 0.2736842105263158, 'psh_ratio': 0.14736842105263157, 'ack_ratio': 0.9473684210526315, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T16:48:22.346276', 'packet_count': 100, 'protocol_distribution': {'tcp': 95, 'udp': 4, 'icmp': 0}, 'connection_info': {'unique_connections': 28, 'unique_src_ips': 22, 'unique_dst_ips': 8}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.

Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.